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Integration

Standard Integrals

Definition: Integration is the reverse of differentiation. The indefinite integral of $f(x)$ is:

$$\int f(x)\,dx = F(x) + c$$

where $F'(x) = f(x)$ and c is the constant of integration.

$f(x)$	$\int f(x)\,dx$	Conditions
x^n	$\dfrac{x^{n+1}}{n+1} + c$	$n \neq -1$
$x^{-1} = \dfrac{1}{x}$	$\ln\lvert x\rvert + c$	$x \neq 0$
e^x	$e^x + c$	
e^{kx}	$\dfrac{1}{k}e^{kx} + c$	
$\sin x$	$-\cos x + c$	
$\cos x$	$\sin x + c$	
$\sec^2 x$	$\tan x + c$	
$\sin(kx)$	$-\dfrac{1}{k}\cos(kx) + c$	
$\cos(kx)$	$\dfrac{1}{k}\sin(kx) + c$	

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$\dfrac{f'(x)}{f(x)}$	$\ln\lvert f(x)\rvert + c$	Key recognition pattern

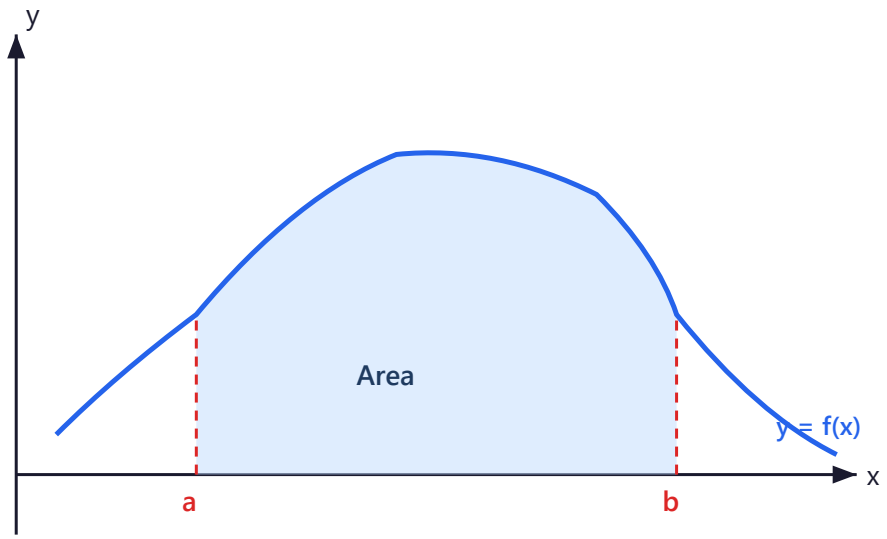
Exam Tip: Always include $+c$ for indefinite integrals. Forgetting the constant of integration is a common mark lost in exams.

Definite Integrals and Area

The Fundamental Theorem of Calculus:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

where $F(x)$ is any antiderivative of $f(x)$.



Important area rules:

- Area between the curve and the x-axis: $A = \displaystyle\int_a^b f(x)\,dx$ (when $f(x) \geq 0$)
- If $f(x) < 0$ on an interval, the integral is negative. Take the absolute value for area.
- Area between two curves: $A = \displaystyle\int_a^b \lvert f(x) - g(x)\rvert\,dx$

Worked Example: Area Between Curves

Find the area enclosed between $y = x^2$ and $y = 4x - x^2$.

Solution: Find intersections: $x^2 = 4x - x^2 \implies 2x^2 - 4x = 0 \implies x = 0$ or $x = 2$.

$$A = \int_0^2 (4x - x^2 - x^2)\,dx = \int_0^2 (4x - 2x^2)\,dx$$

$$= \left[2x^2 - \frac{2x^3}{3}\right]_0^2 = 8 - \frac{16}{3} = \frac{8}{3}$$

Integration by Substitution

Substitution Method: To evaluate $\int f(g(x)) \cdot g'(x)\,dx$, let $u = g(x)$, then $du = g'(x)\,dx$:

$$\int f(g(x)) \cdot g'(x)\,dx = \int f(u)\,du$$

Worked Example: Substitution

Find $\displaystyle\int x\sqrt{x^2+1}\,dx$.

Solution: Let $u = x^2 + 1$, so $\dfrac{du}{dx} = 2x$, giving $x\,dx = \frac{1}{2}\,du$.

$$\int x\sqrt{x^2+1}\,dx = \frac{1}{2}\int u^{1/2}\,du = \frac{1}{2}\cdot\frac{2}{3}u^{3/2}+c = \frac{1}{3}(x^2+1)^{3/2}+c$$

Exam Tip: For definite integrals with substitution, change the limits to the new variable u. This avoids having to substitute back. For example, if $u = x^2+1$ and x goes from 0 to 2, then u goes from 1 to 5.

Integration by Parts

Integration by Parts:

$$\int u\,\frac{dv}{dx}\,dx = uv - \int v\,\frac{du}{dx}\,dx$$

Choose u using LIATE priority: Logs, Inverse trig, Algebraic, Trig, Exponentials.

Worked Example: Integration by Parts

Find $\int xe^x\,dx$.

Solution: Let $u = x$ (algebraic) and $\frac{dv}{dx} = e^x$, so $\frac{du}{dx} = 1$ and $v = e^x$.

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + c = e^x(x-1)+c$$

Worked Example: Integration by Parts (Repeated)

Find $\int \ln x \, dx$.

Solution: Let $u = \ln x$ and $\frac{dv}{dx} = 1$, so $\frac{du}{dx} = \frac{1}{x}$ and $v = x$.

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - x + c$$

Integration Using Partial Fractions

When integrating rational functions, decompose into partial fractions first:

Worked Example: Partial Fractions

Find $\int \frac{3x+5}{(x+1)(x+2)} \, dx$.

Solution: Decompose: $\frac{3x+5}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$.

Cover-up: $x = -1$: $A = \frac{2}{1} = 2$. $x = -2$: $B = \frac{-1}{-1} = 1$.

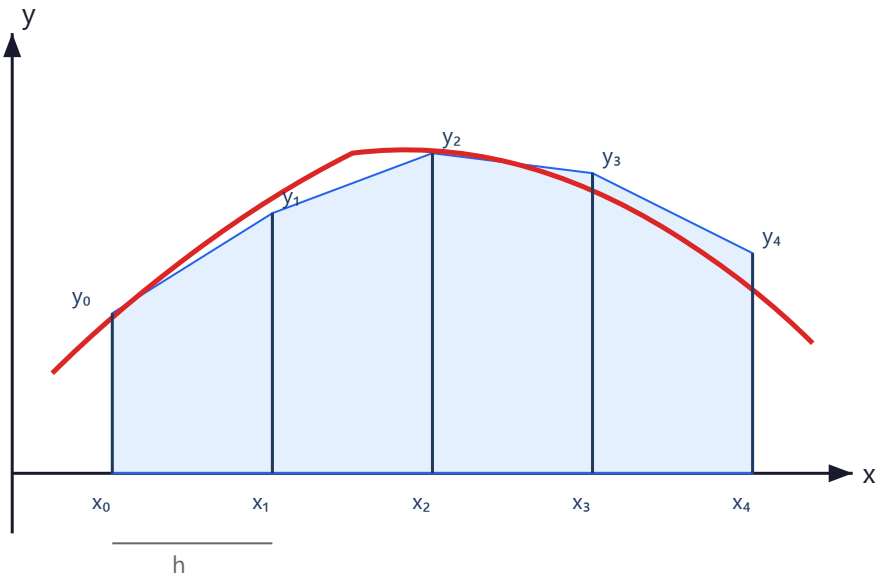
$$\int \frac{3x+5}{(x+1)(x+2)} \, dx = \int \frac{2}{x+1} \, dx + \int \frac{1}{x+2} \, dx = 2 \ln |x+1| + \ln |x+2| + c$$

Trapezium Rule

Trapezium Rule:

$$\int_a^b y \, dx \approx \frac{h}{2} \left[y_0 + y_n + 2(y_1 + y_2 + \cdots + y_{n-1}) \right]$$

where $h = \dfrac{b-a}{n}$ is the strip width and n is the number of strips.



Worked Example: Trapezium Rule

Use the trapezium rule with 4 strips to estimate $\displaystyle\int_0^2 e^{x^2}\,dx$.

Solution: $h = \dfrac{2-0}{4} = 0.5$. Calculate ordinates:

x	0	0.5	1	1.5	2
e^{x^2}	1	1.2840	2.7183	9.4877	54.598

$$\approx \frac{0.5}{2}[1 + 54.598 + 2(1.2840 + 2.7183 + 9.4877)]$$

$$= 0.25[55.598 + 2(13.490)] = 0.25 \times 82.578 = 20.6 \text{ (3 s.f.)}$$

Differential Equations (Separation of Variables)

Separation of Variables: For a differential equation of the form $\frac{dy}{dx} = f(x)g(y)$:

$$\int \frac{1}{g(y)}\,dy = \int f(x)\,dx$$

Worked Example: Separation of Variables

Solve $\frac{dy}{dx} = 3x^2y$, given $y = 2$ when $x = 0$.

Solution: Separate variables:

$$\int \frac{1}{y}\,dy = \int 3x^2\,dx$$

$$\ln|y| = x^3 + c$$

Using the initial condition $y = 2, x = 0$: $\ln 2 = 0 + c$, so $c = \ln 2$.

$$\ln|y| = x^3 + \ln 2 \implies y = 2e^{x^3}$$

Exam Tip: When solving differential equations, always use the given boundary condition to find the constant of integration. Present your final answer as $y = \ldots$ if possible, and check by substituting back into the original equation.

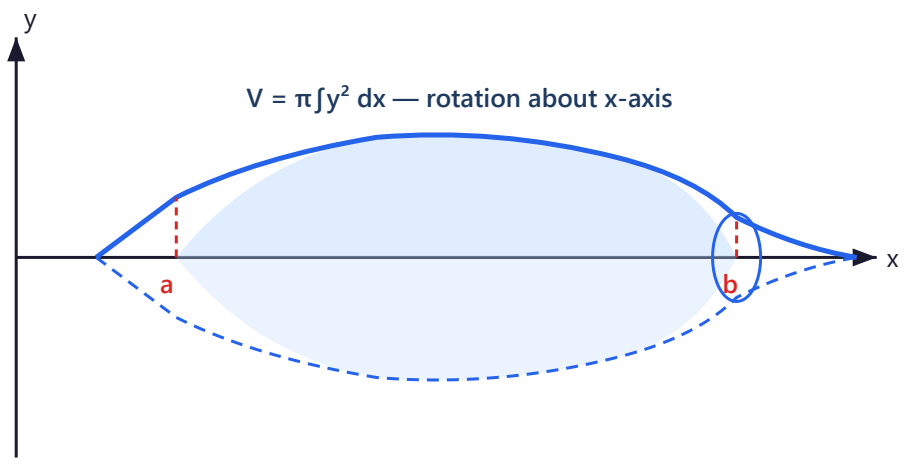
Volumes of Revolution

Rotation about the x-axis: The volume formed when the region between $y = f(x)$, $x = a$ and $x = b$ is rotated 360° about the x-axis:

$$V = \pi \int_a^b y^2 \, dx$$

Rotation about the y-axis:

$$V = \pi \int_c^d x^2 \, dy$$



Worked Example: Volume of Revolution

The region bounded by $y = \sqrt{x}$, $x = 1$ and $x = 4$ is rotated 360° about the x-axis. Find the volume.

Solution: $y = \sqrt{x} \implies y^2 = x$.

$$V = \pi \int_1^4 x \, dx = \pi \left[\frac{x^2}{2} \right]_1^4 = \pi \left(8 - \frac{1}{2} \right) = \frac{15\pi}{2}$$

Worked Example: Volume with Subtraction

The region bounded by $y = x^2 + 1$ and $y = 5$ is rotated 360° about the y-axis. Find the volume.

Solution: From $y = x^2 + 1$: $x^2 = y - 1$. Limits in y: curve meets $y = 5$ at $x^2 = 4$, and meets y-axis at $y = 1$.

$$V = \pi \int_1^5 (y-1)\,dy = \pi \left[\frac{y^2}{2} - y\right]_1^5 = \pi\left[\left(\frac{25}{2} - 5\right) - \left(\frac{1}{2} - 1\right)\right] = 8\pi$$

Exam Tip: Always include π in volume answers and leave in exact form. The formula is $\pi \int y^2\,dx$ — square the full expression for y before integrating. A common error is integrating y then squaring the result.

Integrating Trigonometric Functions

Use double angle identities to integrate even powers of sine and cosine:

$$\cos^2 x = \frac{1 + \cos 2x}{2}, \qquad \sin^2 x = \frac{1 - \cos 2x}{2}$$

Worked Example: Integrating $\cos^2 x$

Find $\displaystyle\int \cos^2 x\,dx$.

Solution: Substitute the identity $\cos^2 x = \dfrac{1 + \cos 2x}{2}$:

$$\int \cos^2 x \, dx = \int \frac{1 + \cos 2x}{2} \, dx = \frac{x}{2} + \frac{\sin 2x}{4} + c$$

Worked Example: Definite Integral with $\sin^2 x$

Evaluate $\int_0^{\pi} \sin^2 x \, dx$.

Solution: Use $\sin^2 x = \dfrac{1 - \cos 2x}{2}$:

$$\int_0^{\pi} \frac{1 - \cos 2x}{2} \, dx = \left[\frac{x}{2} - \frac{\sin 2x}{4} \right]_0^{\pi} = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

Exam Tip: Never try to integrate $\sin^2 x$ or $\cos^2 x$ directly — always use the double angle identity first. These appear frequently in definite integral and volume of revolution questions.

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